

Signed sub-optimality bound for sample-path AIS

Extended Theorem 3

Lemma 1 (Policy evaluation). *Let $\hat{\pi} = (\hat{\pi}_1, \dots, \hat{\pi}_T)$, $\hat{\pi}_t : \mathcal{Z} \rightarrow \mathcal{A}$, be any AIS policy and $\pi_t = \hat{\pi}_t \circ \sigma$, $\pi_t : \mathcal{H} \rightarrow \mathcal{A}$ its lifted version with $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$ for all $t \in \{1, \dots, T\}$. Define*

$$\alpha_t^{\hat{\pi}}(h_t) = V_t^{\pi}(h_t) - \hat{V}^{\hat{\pi}}(\sigma_t(h_t)), \quad (1)$$

$$\beta_t^{\hat{\pi}}(h_t, a_t) = Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t), \quad (2)$$

and

$$\Delta_t^{\hat{\pi}}(h_t, a_t) = \left[c_t(h_t, a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) \right] \quad (3)$$

$$- \left[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1} | h_t, a_t) \right]. \quad (4)$$

Then, for all $h_t \in \mathcal{H}_t$ and $a_t \in \mathcal{A}$, the following holds

$$\beta_t^{\hat{\pi}}(h_t, a_t) = \Delta_t^{\hat{\pi}}(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (5)$$

$$\alpha_t^{\hat{\pi}}(h_t) = \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))). \quad (6)$$

where $\alpha_{T+1}^{\hat{\pi}} \equiv 0$.

Proof. We first establish the recursion (5) and then the identity (6).

Step 1 (the $\beta^{\hat{\pi}}$ -recursion). Fix $t \in \{1, \dots, T\}$, $h_t \in \mathcal{H}_t$, and $a_t \in \mathcal{A}$. Adding and subtracting $\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t)$, and using $\hat{P}_t(\cdot | \sigma_t(h_t), a_t) = \hat{P}_t^{\sigma}(\cdot | h_t, a_t)$, we get

$$\begin{aligned} \beta_t^{\hat{\pi}}(h_t, a_t) &= Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t) \\ &= [c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] \\ &\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t(dz_{t+1} | \sigma_t(h_t), a_t) \\ &= \underbrace{[c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1} | h_t, a_t)}_{\Delta_t^{\hat{\pi}}(h_t, a_t)} \\ &\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t), \end{aligned}$$

where the first bracket is $\Delta_t^{\hat{\pi}}(h_t, a_t)$ by (3). For the remaining terms, we have

$$\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) = \int \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) P_{Y,t}(dy_{t+1} | h_t, a_t).$$

Thus the last term can be written as

$$\begin{aligned} & \int V_{t+1}^\pi(h_{t+1})P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ & - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1})P_t^\sigma(dz_{t+1} \mid h_t, a_t) = \int \left[V_{t+1}^\pi(h_{t+1}) - \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) \right] P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ & = \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1})P_{Y,t}(dy_{t+1} \mid h_t, a_t) \end{aligned}$$

so the two integrals against $P_{Y,t}$ combine into

$$\int \left[V_{t+1}^\pi(h_{t+1}) - \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) \right] P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1})P_{Y,t}(dy_{t+1} \mid h_t, a_t),$$

This establishes (5). At $t = T$, we have $\alpha_{T+1}^{\hat{\pi}} \equiv 0$ and $\Delta_T^{\hat{\pi}}(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$, so (5) reads $\beta_T^{\hat{\pi}} = \Delta_T^{\hat{\pi}}$.

Step 2 (the $\alpha^{\hat{\pi}}$ -identity). Since $\hat{\pi}_t$ is deterministic and $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$, the value functions are the Q -functions evaluated at the prescribed actions:

$$V_t^\pi(h_t) = Q_t^\pi(h_t, \hat{\pi}_t(\sigma_t(h_t))), \quad \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) = \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))).$$

Then, we can write

$$\alpha_t^{\hat{\pi}^*}(h_t) = V_t^\pi(h_t) - \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) \tag{7}$$

$$= Q_t^\pi(h_t, \hat{\pi}_t(\sigma_t(h_t))) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))) \tag{8}$$

$$= \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))) \tag{9}$$

□

Lemma 2 (Optimality). *Define*

$$\alpha_t^*(h_t) = V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \tag{10}$$

$$\beta_t^*(h_t, a_t) = Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t), \tag{11}$$

and

$$\Delta_t(h_t, a_t) = \left[c_t(h_t, a_t) + \int \hat{V}_{t+1}^*(z_{t+1})P_t^\sigma(dz_{t+1} \mid h_t, a_t) \right] \tag{12}$$

$$- \left[\hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^*(z_{t+1})\hat{P}_t^\sigma(dz_{t+1} \mid h_t, a_t) \right]. \tag{13}$$

Then, the following holds

$$\beta_t^*(h_t, a_t) = \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1})P_{Y,t}(dy_{t+1} \mid h_t, a_t), \tag{14}$$

$$\beta_t^*(h_t, \pi_t^*(h_t)) \leq \alpha_t^*(h_t) \leq \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \tag{15}$$

where $\alpha_{T+1}^* \equiv 0$.

Proof. Step 1 (the β^ -recursion).* [Complete Step 1, similar to Lemma 1.1]

The derivation is identical to Step 1 in the proof of Lemma 1 with $(Q_t^\pi, \hat{Q}_t^{\hat{\pi}}, V_{t+1}^\pi, \hat{V}_{t+1}^{\hat{\pi}})$ replaced by $(Q_t^*, \hat{Q}_t^*, V_{t+1}^*, \hat{V}_{t+1}^*)$, add and subtract $\int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t)$, and identify the cost-and-kernel-mismatch bracket as $\Delta_t(h_t, a_t)$ from (12).

Step 2 (the α^ -inequalities).* The two inequalities in (15) follow from the two optimality properties, used one at a time.

Upper bound. By optimality of $\hat{\pi}_t^*$ on the AIS side, $\hat{V}_t^*(\sigma_t(h_t)) = \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t)))$, while on the true side $V_t^*(h_t) = \inf_{a \in \mathcal{A}} Q_t^*(h_t, a) \leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t)))$. Therefore,

$$\begin{aligned} \alpha_t^*(h_t) &= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ &\leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) = \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \end{aligned}$$

Lower bound. By optimality of π_t^* on the true side, $V_t^*(h_t) = Q_t^*(h_t, \pi_t^*(h_t))$, while on the AIS side $\hat{V}_t^*(\sigma_t(h_t)) = \inf_{a \in \mathcal{A}} \hat{Q}_t^*(\sigma_t(h_t), a) \leq \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t))$. Therefore,

$$\begin{aligned} \alpha_t^*(h_t) &= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ &\geq Q_t^*(h_t, \pi_t^*(h_t)) - \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t)) = \beta_t^*(h_t, \pi_t^*(h_t)). \end{aligned}$$

Combining the two gives (15). $\square$

Theorem 1. Suppose that for each $t \in \{1, \dots, T\}$, there exist measurable functions $\alpha_t^{\hat{\pi}^*}: \mathcal{H}_t \rightarrow \mathbb{R}$, $\alpha_t^*: \mathcal{H}_t \rightarrow \mathbb{R}$, $\beta_t^{\hat{\pi}^*}: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$, and $\beta_t^*: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ such that

$$\beta_T^{\hat{\pi}^*}(h_T, a_T) \geq \Delta_T(h_T, a_T) \geq \beta_T^*(h_T, a_T) \quad (16)$$

and for $t < T$:

$$\beta_t^{\hat{\pi}^*}(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (17)$$

$$\beta_t^*(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (18)$$

and for all $t \in \{1, \dots, T\}$:

$$\alpha_t^{\hat{\pi}^*}(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (19)$$

$$\alpha_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)), \quad (20)$$

where Δ_t is as defined in (12) (in particular, at terminal time, $\Delta_T(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$), and π_t^* and $\hat{\pi}_t^*$ are as in Lemma 2. Then, for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$, the AIS-based policy $\pi_t(h_t) = \hat{\pi}_t^*(\sigma_t(h_t))$ satisfies

$$V_t^\pi(h_t) - V_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t). \quad (21)$$

Proof. Throughout the proof, let $\tilde{\alpha}_t^{\hat{\pi}^*}$, $\tilde{\beta}_t^{\hat{\pi}^*}$, $\tilde{\alpha}_t^*$, and $\tilde{\beta}_t^*$ denote the *exact* signed error functions of Lemmas 1 and 2:

$$\tilde{\alpha}_t^{\hat{\pi}^*}(h_t) = V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \quad \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t) = Q_t^\pi(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 1 specialized to $\hat{\pi} = \hat{\pi}^*$, and

$$\tilde{\alpha}_t^*(h_t) = V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \quad \tilde{\beta}_t^*(h_t, a_t) = Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 2. Regarding the specialization $\hat{\pi} = \hat{\pi}^*$ in Lemma 1: since $\hat{\pi}_t^*(z_t)$ is a minimizer of $\hat{Q}_t^*(z_t, \cdot)$, evaluating the AIS dynamic program along $\hat{\pi}^*$ shows that the policy-evaluation value functions of $\hat{\pi}^*$ coincide with the optimal ones, $\hat{V}_t^{\hat{\pi}^*} = \hat{V}_t^*$ and $\hat{Q}_t^{\hat{\pi}^*} = \hat{Q}_t^*$ for all t ; consequently, the integrand $\hat{V}_{t+1}^{\hat{\pi}^*}$ in (??) equals $\hat{V}_{t+1}^*$ and the mismatch terms coincide, $\Delta_t^{\hat{\pi}^*} = \Delta_t$, the same Δ_t as in Lemma 2 and in the statement of the theorem. Thus Lemmas 1 and 2 give, for all t , h_t , and a_t (with $\tilde{\alpha}_{T+1}^{\hat{\pi}^*} \equiv 0$ and $\tilde{\alpha}_{T+1}^* \equiv 0$):

$$\begin{aligned}\tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\ \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) &= \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))),\end{aligned}\tag{22}$$

$$\begin{aligned}\tilde{\beta}_t^*(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\ \tilde{\beta}_t^*(h_t, \pi_t^*(h_t)) &\leq \tilde{\alpha}_t^*(h_t).\end{aligned}\tag{23}$$

Step 1 (the candidate functions compared to the exact errors). We show by backward induction that for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$,

$$\alpha_t^{\hat{\pi}^*}(h_t) \geq \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) \quad \text{and} \quad \alpha_t^*(h_t) \leq \tilde{\alpha}_t^*(h_t).\tag{24}$$

Base case ($t = T$). Since $\tilde{\alpha}_{T+1}^{\hat{\pi}^*} \equiv 0$ and $\tilde{\alpha}_{T+1}^* \equiv 0$, the recursions in (22) and (23) give $\tilde{\beta}_T^{\hat{\pi}^*} = \tilde{\beta}_T^* = \Delta_T$. Hence, by (16), $\beta_T^{\hat{\pi}^*} \geq \tilde{\beta}_T^{\hat{\pi}^*}$ and $\beta_T^* \leq \tilde{\beta}_T^*$. Therefore, by (19) and the identity in (22),

$$\alpha_T^{\hat{\pi}^*}(h_T) \geq \beta_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) = \tilde{\alpha}_T^{\hat{\pi}^*}(h_T),$$

and, by (20) and the inequality in (23),

$$\alpha_T^*(h_T) \leq \beta_T^*(h_T, \pi_T^*(h_T)) \leq \tilde{\beta}_T^*(h_T, \pi_T^*(h_T)) \leq \tilde{\alpha}_T^*(h_T).$$

Induction step ($t < T$). Suppose (24) holds at $t+1$. Then, by (17), the induction hypothesis, and the recursion in (22),

$$\begin{aligned}\beta_t^{\hat{\pi}^*}(h_t, a_t) &\geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\geq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t),\end{aligned}$$

and then, exactly as in the base case, $\alpha_t^{\hat{\pi}^*}(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t)$. Symmetrically, by (18), the induction hypothesis, and the recursion in (23),

$$\begin{aligned}\beta_t^*(h_t, a_t) &\leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\leq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^*(h_t, a_t),\end{aligned}$$

and then $\alpha_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)) \leq \tilde{\beta}_t^*(h_t, \pi_t^*(h_t)) \leq \tilde{\alpha}_t^*(h_t)$. This establishes (24) for all t .

Step 2 (sub-optimality bound). Adding and subtracting $\hat{V}_t^*(\sigma_t(h_t))$,

$$V_t^\pi(h_t) - V_t^*(h_t) = [V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t))] - [V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t))] = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t).$$

Combining Step 2 with (24),

$$V_t^\pi(h_t) - V_t^*(h_t) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t),$$

which is (21). □

Remark 1. With the hypotheses of Theorem 1 flipped, we can obtain a lower bound of a symmetric form. Concretely, suppose that for each t there exist measurable functions $\underline{\alpha}_t^{\hat{\pi}^*}, \underline{\alpha}_t^* : \mathcal{H}_t \rightarrow \mathbb{R}$ and $\underline{\beta}_t^{\hat{\pi}^*}, \underline{\beta}_t^* : \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ such that

$$\underline{\beta}_T^{\hat{\pi}^*}(h_T, a_T) \leq \Delta_T(h_T, a_T) \leq \underline{\beta}_T^*(h_T, a_T), \quad (25)$$

$$\underline{\beta}_t^{\hat{\pi}^*}(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^{\hat{\pi}^*} dP_{Y,t}, \quad \underline{\beta}_t^*(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^* dP_{Y,t}, \quad (26)$$

$$\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \underline{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad \underline{\alpha}_t^*(h_t) \geq \underline{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (27)$$

i.e. the conditions (16)–(20) with every inequality reversed and the $\star$ -anchor (20) moved from $\pi_t^*(h_t)$ to $\hat{\pi}_t^*(\sigma_t(h_t))$. The same induction as in the proof of Theorem 1, with the inequalities reversed, then yields

$$V_t^\pi(h_t) - V_t^*(h_t) \geq \underline{\alpha}_t^{\hat{\pi}^*}(h_t) - \underline{\alpha}_t^*(h_t). \quad (28)$$

This lower bound is not informative. The quantity $\underline{\alpha}_t^{\hat{\pi}^*} - \underline{\alpha}_t^*$ is guaranteed to be non-positive, so (28) can never improve on the trivial bound $V_t^\pi - V_t^* \geq 0$. The reason is that the $\star$ -anchor in (27) is forced to $\hat{\pi}_t^*(\sigma_t(h_t))$, which is exactly the action that anchors the policy-evaluation stream in the first inequality of (27). The two streams then satisfy the identical recursion (26), with the same mismatch Δ_t , the same kernel $P_{Y,t}$, and the same terminal value Δ_T from (25); hence the tightest admissible choices of $\underline{\alpha}_t^{\hat{\pi}^*}$ and $\underline{\alpha}_t^*$ coincide, giving $\underline{\alpha}_t^{\hat{\pi}^*} - \underline{\alpha}_t^* \leq 0$. We therefore omit the lower bound from the statement of Theorem 1.

Corollary 1 (Recovery of the absolute-value bound). *Let $\alpha_t : \mathcal{H}_t \rightarrow \mathbb{R}_{\geq 0}$ and $\beta_t : \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ be measurable functions such that*

$$\beta_T(h_T, a_T) \geq |\Delta_T(h_T, a_T)|, \quad (29)$$

and for $t < T$,

$$\beta_t(h_t, a_t) \geq |\Delta_t(h_t, a_t)| + \int \alpha_{t+1}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (30)$$

and for all $t \in \{1, \dots, T\}$,

$$\alpha_t(h_t) \geq \max \left\{ \beta_t(h_t, \pi_t^*(h_t)), \beta_t(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \right\}. \quad (31)$$

Then the AIS-based policy $\pi_t(h_t) = \hat{\pi}_t^(\sigma_t(h_t))$ satisfies, for all $t \in \{1, \dots, T\}$ and all $h_t \in \mathcal{H}_t$,*

$$|V_t^\pi(h_t) - V_t^*(h_t)| \leq 2\alpha_t(h_t). \quad (32)$$

Proof. Let α_t and β_t satisfy (29)–(31), and choose the functions in Theorem 1 as

$$\alpha_t^{\hat{\pi}^*} = \alpha_t, \quad \alpha_t^* = -\alpha_t, \quad \beta_t^{\hat{\pi}^*} = \beta_t, \quad \beta_t^* = -\beta_t. \quad (33)$$

We show that this choice satisfies the hypotheses of Theorem 1, and then use its result to obtain (32).

Terminal condition (16). The hypothesis (29) states that $\beta_T \geq |\Delta_T|$, which implies $\beta_T \geq \Delta_T$ and $\beta_T \geq -\Delta_T$. Under (33), the first is $\beta_T^{\hat{\pi}^*} \geq \Delta_T$ and the second, negated, is $-\beta_T \leq \Delta_T$, i.e. $\beta_T^* \leq \Delta_T$. Together these are the terminal condition $\beta_T^{\hat{\pi}^*} \geq \Delta_T \geq \beta_T^*$ of (16).

Recursions (17) and (18). Writing $I_t := \int \alpha_{t+1}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t)$ for the integral term, the hypothesis (30) states that $\beta_t \geq |\Delta_t| + I_t$. Since $|\Delta_t| \geq \Delta_t$ and $|\Delta_t| \geq -\Delta_t$, this implies both

$$\beta_t(h_t, a_t) \geq \Delta_t(h_t, a_t) + I_t \quad \text{and} \quad \beta_t(h_t, a_t) \geq -\Delta_t(h_t, a_t) + I_t. \quad (34)$$

Under (33), the first inequality in (34) is exactly the policy-evaluation recursion (17), since $\beta_t^{\hat{\pi}^*} = \beta_t$ and $\alpha_{t+1}^{\hat{\pi}^*} = \alpha_{t+1}$. For the optimality recursion (18), negating the second inequality in (34) gives $-\beta_t \leq \Delta_t - I_t$, and substituting $\beta_t^* = -\beta_t$ and $\alpha_{t+1}^* = -\alpha_{t+1}$ yields

$$\beta_t^*(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t),$$

which is (18). Hence (30) implies (17) and (18) of Theorem 1.

Recursions (19) and (20). The hypothesis (31) states that $\alpha_t \geq \max\{\beta_t(\cdot, \pi_t^*), \beta_t(\cdot, \hat{\pi}_t^* \circ \sigma_t)\}$, which implies $\alpha_t \geq \beta_t(\cdot, \hat{\pi}_t^* \circ \sigma_t)$ and $\alpha_t \geq \beta_t(\cdot, \pi_t^*)$. Under (33), the first is the policy-evaluation anchor $\alpha_t^{\hat{\pi}^*} \geq \beta_t^{\hat{\pi}^*}(\cdot, \hat{\pi}_t^* \circ \sigma_t)$ of (19), and the second, negated, is $-\alpha_t \leq -\beta_t(\cdot, \pi_t^*)$, i.e. $\alpha_t^* \leq \beta_t^*(\cdot, \pi_t^*)$, which is (20).

We may now invoke Theorem 1. The functions $(\alpha_t^{\hat{\pi}^*}, \alpha_t^*, \beta_t^{\hat{\pi}^*}, \beta_t^*) = (\alpha_t, -\alpha_t, \beta_t, -\beta_t)$ satisfy its hypotheses (16)–(20), and (21) therefore yields

$$V_t^\pi(h_t) - V_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t) = \alpha_t(h_t) - (-\alpha_t(h_t)) = 2\alpha_t(h_t). \quad (35)$$

For the matching lower side, by definition of the optimal value function, for any policy π , we have $V_t^\pi(h_t) \geq V_t^*(h_t)$, and since $\alpha_t \geq 0$,

$$V_t^\pi(h_t) - V_t^*(h_t) \geq 0 \geq -2\alpha_t(h_t). \quad (36)$$

Combining (35) and (36) gives $-2\alpha_t(h_t) \leq V_t^\pi(h_t) - V_t^*(h_t) \leq 2\alpha_t(h_t)$, which is (32). $\square$